
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

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Synopsis

On

NEUROSCRIBE

1) INTRODUCTION

1.1 OVERVIEW

In today's digital era, a vast amount of medical information is available in the form of reports, research articles, and health records. However, this information is often complex and difficult for patients and non-experts to understand, creating a communication gap between healthcare professionals and users.

To address this issue, this project introduces **NeuroScribe**, an LLM-based system that summarizes and simplifies medical text. It uses an agent-based approach with a Medical Reader Agent, Summarizer Agent, and Simplifier Agent to convert lengthy and technical content into short, easy-to-understand explanations.

The system processes different types of medical data and provides both technical summaries and simplified versions. Overall, NeuroScribe aims to improve understanding, accessibility, and communication in the healthcare domain.

1.2 PURPOSE

The purpose of this project is to improve the accessibility and understanding of medical information by:

- Providing simplified explanations of complex medical texts for non-expert users.
- Generating concise summaries of lengthy medical reports and research articles
- Bridging the communication gap between healthcare professionals and patients
- Helping users easily understand medical terminology and health conditions.
- Enabling quick and efficient interpretation of medical documents.
- Supporting informed decision-making through clear and accurate information.
- Contributing to better healthcare communication by leveraging LLMs and NLP technologies.

2) LITERATURE SURVEY

The literature survey focuses on analysing existing systems developed for medical text processing, particularly in the areas of text summarization and language simplification. Various research works and tools related to Natural Language Processing (NLP), medical document analysis, and Large Language Models (LLMs) have been studied to understand their capabilities and limitations. The survey highlights the need for a system that not only summarizes medical content but also makes it understandable for non-expert users.

2.1 EXISTING PROBLEM

- Most existing systems generate technical summaries but do not simplify medical language for patients or general users.
- Many tools focus only on summarization and ignore the need for easy-to-understand explanations.
- Medical texts contain complex terminology, making them difficult for non-medical users to interpret.
- Existing simplification tools often lack contextual understanding, leading to inaccurate or incomplete explanations.
- There is a lack of integrated systems that combine both summarization and simplification in a single platform.
- Many solutions are not user-friendly and fail to address real-world patient communication needs.
- Due to these limitations, patients and general users still face difficulty in understanding medical information clearly.

2.2 PROPOSED SOLUTION

- The proposed system, **NeuroScribe**, is an LLM-based platform for medical text summarization and simplification.
- It focuses on converting complex medical data into concise summaries and simple explanations.
- The system uses an agent-based architecture including Medical Reader, Summarizer, and Simplifier agents.
- It supports processing of various medical datasets such as reports, articles, and health records.
- It provides dual outputs: technical summaries and patient-friendly simplified content.
- The platform improves accessibility and understanding of medical information for non-expert users.
- The system is scalable, efficient, and can be integrated into healthcare platforms and applications.

3) THEORETICAL ANALYSIS

3.1 BLOCK DIAGRAM

SIMPLIFIED NEUROSCRIBE WORKFLOW

How We Make Medical Data Easy to Understand

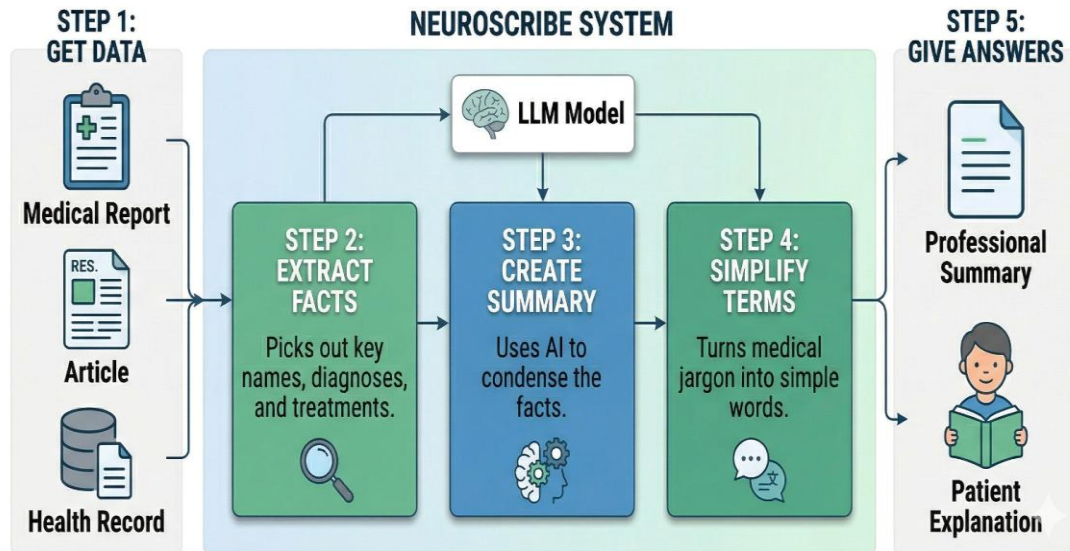


Fig1. Block Diagram

3.2 HARDWARE / SOFTWARE DESIGNING

The development of **NeuroScribe** is based on a modern, scalable, and AI-driven technology stack that ensures efficient processing of medical text and seamless user interaction.

- **Frontend Layer:** Implemented using React.js along with HTML, CSS, and JavaScript to provide a responsive, interactive, and user-friendly interface for input and output visualization.
- **Backend Layer:** Developed using Python and FastAPI to handle API requests, manage workflows, and integrate AI models efficiently.
- **AI Processing Layer:** Utilizes advanced NLP frameworks such as LangChain and LangGraph for agent-based workflow management, along with Hugging Face Transformers for text summarization and simplification tasks.
- **Cloud Integration:** OpenAI API is used to leverage powerful Large Language Models for generating accurate summaries and human-friendly explanations.

4) APPLICATIONS

The proposed system, **NeuroScribe**, can be applied in various domains to improve the understanding and accessibility of medical information for different types of users.

- **Healthcare Sector:** Helps patients understand medical reports, prescriptions, and diagnoses in simple language.
- **Hospitals and Clinics:** Assists doctors and staff in quickly summarizing patient records and clinical data.
- **Medical Education:** Supports students in understanding complex medical concepts through simplified explanations.
- **Telemedicine Platforms:** Enhances communication between doctors and patients during online consultations.
- **Health Awareness Platforms:** Provides easy-to-understand medical content for public awareness and education.
- **Research and Documentation:** Helps researchers summarize lengthy medical articles and papers efficiently.
- **Personal Health Management:** Enables individuals to better understand their health conditions and reports.
- **Government and Healthcare Initiatives:** Supports digital health programs by making information more accessible and user-friendly.

5) REFERENCES

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